

Harsh Desai

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Experimental systems engineer in low-pressure plasma, vacuum/HV test equipment, automation, detector calibration, and ADC/FPGA readout; targeting semiconductor process, equipment, and diagnostics roles.

EDUCATION

University of Michigan

Master of Engineering, Space Systems Engineering | GPA: 3.79/4.0

Vellore Institute of Technology (VIT)

Bachelor of Technology, Mechanical Engineering | GPA: 7.78/10

Ann Arbor, MI, USA

Aug 2024–Dec 2025

Vellore, India

Jul 2019–Jul 2023

TECHNICAL SKILLS

Plasma, Vacuum & Process Equipment: Low-pressure hollow-cathode plasma operation, 1.3 kV HV interface validation, vacuum chamber operation (CTI-Cryogenics Hi-Vac; roughing pump operated at 1 mTorr), RPA, emissive-probe and Langmuir-probe diagnostics, CEM/ESA calibration

Automation, Controls & DAQ: Python, LabVIEW (CLAD prep), MATLAB (learning), GPIB/IEEE-488, USB-serial, serial data logging, NI VISA (PyVISA, PyMeasure), Keithley DAQs and SMUs, TDK Lambda PSUs, High Voltage Switching Operations

FPGA, Digital Systems & Modeling: Verilog (learning), VHDL (learning), SIMION, SRIM, Amptek DPPMCA, FPGA Design Workflow, AMD Vivado, MATLAB HDL Coder & DSP Toolbox, SPENVIS, SWMF & Magnetosphere-Ionosphere Models (MAGE, SWMF, HYPERS), Google Data Studio

Mechanical & Manufacturing Tools: ANSYS (GUI/TUI): Fluent, Mechanical; PyANSYS, OptiSlang, OpenFOAM, SolidWorks, Fusion 360, AutoCAD, Autodesk Inventor, SpaceClaim, MIDO

RESEARCH & ENGINEERING EXPERIENCE

Climate and Space Sciences and Engineering, University of Michigan

Ann Arbor, MI, USA

Research Assistant, LVACCS Testing Rig Characterization

Feb 2026–Present

- Validated a 1300 V hollow-cathode plasma-source test workflow by combining HV discharge-box FMEA, Python/PyVISA ignition control, PSU logging, and DAQ synchronization; cut manual steps 15→5 and achieved 98.6% synchronized data
- Supported electrostatic regolith-lofting experiments using Retarding Potential Analyzer (RPA), emissive-probe, and Langmuir-probe diagnostics to compare plasma conditions across test states
- Characterized remote ignition and post-run processing workflows for Spacecraft Charging Device test readiness, improving experimental repeatability and saving ~15 minutes of processing per run
- Built practical experience with low-pressure discharge operation, ignition stability, chamber diagnostics, and parameterized testing, creating a technical bridge toward plasma etch and thin-film deposition equipment

Space Physics Research Lab, University of Michigan

Ann Arbor, MI, USA

Summer Intern, TestBedz Spacecraft Qualification Platform

May 2025–Jul 2025

- Built a full-stack requirement-flowdown platform for TVAC, vibration, and EMI test submissions, matching spacecraft hardware profiles to facility limits across 6 test-type workflows
- Encoded hardware metadata, facility operating envelopes, and acceptance criteria into machine-readable compatibility checks and traceable test plans for repeatable equipment-qualification workflows

Solar and Heliospheric Research Group, University of Michigan

Ann Arbor, MI, USA

Student Research Assistant, SSD Readout Chain

Jan 2025–Dec 2025

- Designed an SSD readout roadmap around science-driven energy-resolution targets, integrating a Zmod ADC 1410 path with a Zynq-7000 architecture to evaluate a 1 MSPS→125 MSPS digitization upgrade
- Verified MATLAB HDL Coder golden models against Vivado co-simulations and diagnosed 125 MHz post-synthesis timing limits, including Worst Negative Slack constraints in the Zynq SoC signal path

L3Harris and University of Michigan

Ann Arbor, MI, USA

Communications Lead, Uranian Orbiter and Probe Mission Study

Sep 2024–May 2025

- Led Uranian mission communications architecture trades and DSN/fault-tolerance requirement flowdown, managing 15 traceable requirements and supporting a 16% (\$450M) cost reduction through an orbiter-only configuration trade

SELECTED INSTRUMENTATION & PROCESS-RELEVANT PROJECTS

Space Instrumentation Calibration & Ion-Optics Series

Jul 2025–Dec 2025

Graduate Course: SPACE 571 | Supervisor: Prof. Stefano Livi

- Calibrated Channel Electron Multiplier response by mapping operating bias, pulse-height distributions, and amplifier gain behavior, connecting detector settings to usable signal quality and count-rate interpretation
- Calibrated the signal chain from charge injection through Charge-Shaper Amplifier, pulse shaping, and MCA acquisition, extracting gain, centroid, FWHM, and bias-dependent count response
- Modeled Electrostatic Analyzer and ion-optics performance using lab energy-angle scans, SIMION, and SRIM to quantify transmission, angular acceptance, energy resolution, FWHM, and filter response

PROJECTS & LEADERSHIP

CANSAT 2022 (NASA & AAS) | Ranked 7th worldwide out of 42 teams

Aug 2021–Jul 2022

- Built a 10 m tether-deployment, payload release, and camera-stabilization system using a DC motor, worm gear, torsion-spring latch, and dual-servo gimbal for controlled descent and fixed 45° imaging

Spaceport America Cup / IREC | Placed 23rd globally and 5th in Asia-Pacific

Apr 2020–Jul 2021

- Launched a Mach 0.9 solid-motor rocket to 10,000 ft and developed an SRAD trajectory simulator using CFD-backed aerodynamic data for flight prediction

CANSAT 2021 (NASA & AAS) | Placed 13th globally and 7th in Asia-Pacific

Aug 2020–Jul 2021

- Modeled maple-seed mono-wing payload descent using Blade Element Theory and transient CFD, with MATLAB telemetry software for real-time dual-payload deployment monitoring